

✂️ Recommended Adjustments

1. Simplify the Heatmap Spatial Index

- **The Plan says:** Use an advanced spatial index of coordinate nodes to track hover on the HTML5 canvas.
- **The Reality Check:** Writing a quadtree or massive coordinate lookup array in Vanilla JS can quickly become a debugging headache.
- **The Fix:** Since your 2D geometry is static and custom-drawn via Bezier curves, pre-define **3 to 5 discrete "Hotspot Zones"** (e.g., Parent Artery, Aneurysm Neck, Aneurysm Sac Dome) using basic bounding boxes or radial distances from a center point. When the mouse enters that radius, pull data from that zone. This achieves the *exact* same high-fidelity tooltip effect with 10% of the math.

2. Safeguard the Canvas Textures during Parameter Toggles

- **The Plan says:** Switch parameter toggles (TAWSS vs OSI) and shift canvas colors immediately.
- **The Reality Check:** Redrawing complex gradients dynamically on a single canvas can sometimes cause visual flickering.
- **The Fix:** Write a single, clean draw function in `app.js` that takes an array of normalized values based on the selected mode. Better yet, use a standard linear interpolation (`lerp`) function to map your metric values smoothly to a color gradient array (e.g., mapping `$0.0 \rightarrow 1.0$` smoothly between `#1F5F99` and `#B83232`).

3. Bulletproof the PDF @media print

- **The Plan says:** Rely strictly on CSS `@media print` over the open dashboard view.
- **The Reality Check:** Browsers are notoriously finicky when printing HTML5 Canvas elements natively via window print commands—sometimes the canvas grids clip or render blank.
- **The Fix:** In your PDF Printer script within `app.js`, right before triggering `window.print()`, use `canvas.toDataURL()` to snapshot your interactive heatmap canvas into a static `<img>` tag inside your report modal. This guarantees that whatever the user sees on the screen perfectly prints onto the PDF document layout without getting cut off.